Matrix Basics: A Short Study Guide

1. What is a matrix?

A matrix is a rectangular array of numbers arranged in rows and columns. The numbers are called the entries or elements of the matrix.

A matrix with m rows and n columns has order m × n (read "m by n"). For example, a matrix with 2 rows and 3 columns has order 2 × 3. The entry in row i and column j is written a_ij.

In this guide a matrix is written on one line with the rows separated by semicolons. For example, [1 2; 3 4] is the matrix whose first row is 1 2 and whose second row is 3 4.

Matrices are used in engineering to describe systems of linear equations, electrical circuits, computer graphics and tables of data.

2. Special types of matrices

A square matrix has the same number of rows and columns. A row matrix has only one row and a column matrix has only one column.

A zero matrix has all of its entries equal to zero. A diagonal matrix is a square matrix whose entries outside the main diagonal are all zero.

The identity matrix, written I, is a square matrix with 1 on the main diagonal and 0 everywhere else. Multiplying any matrix by the identity matrix of the right size leaves it unchanged: AI = IA = A.

A symmetric matrix is a square matrix that is equal to its own transpose, that is A = A^T.

3. Addition and scalar multiplication

Two matrices can be added only if they have the same order. To add them, add the corresponding entries. Subtraction works in the same way. For example, [1 2; 3 4] + [5 6; 7 8] = [6 8; 10 12].

To multiply a matrix by a number (a scalar), multiply every entry by that number. For example, 3 × [1 2; 0 4] = [3 6; 0 12].

Matrix addition is commutative, A + B = B + A, and associative, (A + B) + C = A + (B + C).

4. Matrix multiplication

The product AB is defined only when the number of columns of A equals the number of rows of B. If A has order m × n and B has order n × p, then AB has order m × p.

The entry in row i and column j of AB is found by multiplying the entries of row i of A by the matching entries of column j of B and adding the results.

Example: if A = [1 2; 3 4] and B = [5 6; 7 8], then AB = [19 22; 43 50]. The first entry is 1×5 + 2×7 = 19.

Matrix multiplication is generally not commutative. In most cases AB is not equal to BA. For the same A and B, BA = [23 34; 31 46], which is different from AB.

5. Transpose and determinant

The transpose of a matrix A, written A^T, is obtained by swapping its rows and columns. The entry in row i and column j of A becomes the entry in row j and column i of A^T. If A has order m × n, then A^T has order n × m. For example, if A = [1 2 3; 4 5 6], then A^T = [1 4; 2 5; 3 6].

The determinant of a 2 × 2 matrix [a b; c d] is ad - bc. For example, the determinant of [3 4; 2 5] is 3×5 - 4×2 = 7.

A square matrix whose determinant is zero is called singular. A singular matrix has no inverse.

6. The inverse of a matrix

The inverse of a square matrix A is written A^-1 and satisfies A A^-1 = A^-1 A = I. Only square matrices with a non-zero determinant have an inverse.

For a 2 × 2 matrix [a b; c d], the inverse is 1/(ad - bc) times [d -b; -c a].

Worked example: let A = [4 7; 2 6]. The determinant is 4×6 - 7×2 = 10. So A^-1 = 1/10 × [6 -7; -2 4] = [0.6 -0.7; -0.2 0.4].

Inverses are used to solve systems of linear equations. If AX = B, then X = A^-1 B. A useful rule is (AB)^-1 = B^-1 A^-1.